Maths Week 2: Brackets and equations, written

Year 8 Maths · Algebra · Skill: written method and checking | About 25 minutes, by hand | Name: _____ Date: _____

Rules for today: every answer shows its working, and every equation ends with a **Check** line (put x back in). No check, no tick.

A · Expand and simplify [5 marks]

1. Expand $4(x + 5)$	Answer: _____
2. Expand $3(2x − 7)$	Answer: _____
3. Expand $−2(x − 6)$	Answer: _____
4. Expand and simplify $5(x + 2) + 3(x − 4)$	Answer: _____
5. Expand and simplify $2(4x − 3) − 5(x + 1)$	Answer: _____

B · Solve, with working and a check [8 marks]

6. $3(x + 5) = 36$

x = _____ **Check:** left side = _____ right side = _____

7. $5(x − 2) = 3x + 8$

x = _____ **Check:** left side = _____ right side = _____

8. $3(x + 2) = 2(x + 9)$

x = _____ **Check:** left side = _____ right side = _____

9. $2(3x + 1) − x = 17$

x = _____ **Check:** left side = _____ right side = _____

C · Forming equations [4 marks]

10. I think of a number, add 4, then multiply by 3. The answer is 27. Write an equation and solve it to find my number.

Equation: _____ x = _____ **Check:** _____

11. Three friends each pay £(x + 0.50) for a ticket. The total cost is £4.50. Write an equation and solve it. How much is each ticket?

Equation: _____ x = _____ Each ticket costs £_____ **Check:** _____

D · Error log (3 min)

Look at any question you lost a tick on. Write down what went wrong, in a few words, and the rule that stops it next time. If you got everything right, write one thing the check caught.

Question	What went wrong? (sign / bracket / arithmetic / forgot to check)	My rule for next time

Self-check (5 min)

- ☐ Every equation has a **Check** line and the two sides matched.
- ☐ When a number sat outside a bracket, I multiplied **everything** inside, including its sign.
- ☐ I wrote one line per step, so someone could follow my working.
- ☐ I filled in the error log honestly.

Oral check (5 min, with a parent)

Say it out loud, from memory if you can. Parent asks: **"Why does −4 × −3 become +12 when you expand −4(x − 3)?" (She explains it in her own words.)**

Parent tick: ☐ answered without reading it ☐ answered with a prompt ☐ could not yet

Weekly review (for J, 5 min)

This week's Sparx topic / codes (if different):	
What was weak this week?	
What worked?	
Time taken (HTML / written / total)	
One change for next week	

Parent key and what to look for

A: 1) $4x + 20$ 2) $6x - 21$ 3) $-2x + 12$ 4) $8x - 2$ ($5x + 10 + 3x - 12$) 5) $3x - 11$ ($8x - 6 - 5x - 5$)

B: 6) $x = 7$ (check $3 \times 12 = 36$). 7) $x = 9$ ($5x - 10 = 3x + 8$, $2x = 18$; check $5 \times 7 = 35$ and $3 \times 9 + 8 = 35$). 8) $x = 12$ ($3x + 6 = 2x + 18$; check $3 \times 14 = 42$ and $2 \times 21 = 42$). 9) $x = 3$ ($6x + 2 - x = 17$, $5x = 15$; check $2 \times 10 - 3 = 17$).

C: 10) $3(x + 4) = 27$, $x = 5$ (check $3 \times 9 = 27$). 11) $3(x + 0.50) = 4.50$, $x = 1$, so each ticket costs £1.50 (check $3 \times 1.50 = 4.50$).

Marking: a tick needs the correct answer **and** visible working. A correct answer with no working gets half credit this week. The aim is the habit, not only the number.

Watch for: (a) the second term inside a bracket not being multiplied ($5(x - 2) = 5x - 2$); (b) a dropped sign when the multiplier is negative; (c) a check done on the *working* rather than the *original* equation. The check must use the original equation.

If she gets a check that fails: that is the system working. Ask her to find the first wrong line. That is the same skill as the spot-the-error part on screen.